



Department of Computer Science and Engineering
Premier University

CSE 481: Contemporary Course of Computer Science

Complex Engineering Assignment

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Remarks

Intelligent Smart Transportation and Traffic Management System for Bangladesh

1. Introduction

Bangladesh is one of the most densely populated countries in South Asia, with its metropolitan cities — particularly Dhaka, Chattogram, Sylhet, and Rajshahi — experiencing chronic traffic congestion. The existing traffic control infrastructure relies almost entirely on manual signal operation and static timing cycles that were never designed to handle the sheer volume of vehicles found on today's roads. The result is extended travel times, increased fuel consumption, higher pollution levels, and critically slow emergency response times.

This assignment proposes a comprehensive Intelligent Smart Transportation and Traffic Management System (ISTMS) that brings together Internet of Things (IoT) devices, edge computing hardware, cloud-based analytics, and machine learning to address these longstanding problems. The system is designed to function across both metropolitan and semi-urban areas of Bangladesh, and it is built to scale incrementally as infrastructure and budget allow.

2. Investigation: Current Limitations of Traffic Systems in Bangladesh

2.1 Static Signal Timing

Most traffic signals in Dhaka and Chattogram operate on pre-set, fixed-time cycles that were configured years ago. These timings do not adjust to real-time traffic volumes, which means that during peak hours a busy intersection may still allocate equal time to all directions regardless of actual vehicle density. A road that carries five times more traffic in one direction receives the same signal duration as a quieter road — an inherently inefficient arrangement.

2.2 Absence of Centralized Monitoring

There is no unified command-and-control center that receives live feeds from across a city's road network. Traffic police rely on radio communications and personal observation, which introduces delays and human error. When an incident occurs on one road, nearby intersections cannot automatically adapt to reroute traffic because there is no mechanism for them to receive that information in real time.

2.3 Delayed Emergency Response

Ambulances, fire trucks, and police vehicles frequently get stuck in the same congestion as regular traffic. Without a system to detect approaching emergency vehicles and pre-clear their path by adjusting signals, response times suffer — with potentially fatal consequences for patients and fire victims.

2.4 Lack of Data-Driven Decision Making

No comprehensive historical traffic dataset exists that would allow planners to identify chronic bottlenecks, evaluate the impact of road expansions, or forecast future congestion. Policy decisions are made largely on anecdotal evidence and periodic manual surveys rather than continuous, granular measurement.

2.5 Infrastructure Gaps in Semi-Urban Areas

Outside major cities, road infrastructure, power supply, and internet connectivity are inconsistent. Any proposed solution must account for intermittent network availability and operate with some degree of local intelligence even when cloud connectivity is lost.

3. Proposed System Architecture

The ISTMS is organised across three distinct layers: the IoT sensing layer at ground level, an edge computing layer at the roadside, and a cloud intelligence layer for large-scale analytics and long-term optimisation. These three layers work in concert, with each layer responsible for decisions at the appropriate timescale.

3.1 Layer 1 — IoT Sensing and Data Collection

The sensing layer comprises all physical devices deployed on and around the road network. Each device captures a specific type of data and feeds it upward to the edge layer.

- **Smart Traffic Signal Controllers:** Microcontroller-based units that can receive commands from the edge node and independently switch signal phases. Each controller is equipped with a backup battery to maintain operation during power cuts. Communication is handled over a dedicated 4G LTE link with a Wi-Fi fallback.
- **High-Definition CCTV Cameras:** Installed at intersections and along highway stretches, these cameras capture video at 1080p resolution. On-board image processing chips perform preliminary vehicle detection and counting locally, transmitting only metadata (vehicle count, approximate speed, incident flags) rather than raw video to reduce bandwidth consumption.
- **Inductive Loop Sensors and Radar Units:** Embedded in or above the road surface, these sensors measure vehicle speed, headway, and queue length with high precision. They complement the cameras by providing data even in low-visibility conditions such as rain or fog, which are common during the monsoon season in Bangladesh.
- **GPS-Enabled Public Transport Tracking:** All buses and minibuses operating under city route frameworks will carry a low-cost GPS transponder that transmits location and speed at 10-second intervals. This feeds the cloud model with continuous data on bus bunching and schedule adherence.
- **Environmental Sensors:** Temperature, humidity, and air quality (PM2.5, NO2) sensors are co-located with traffic signal poles. These readings help correlate traffic patterns with weather conditions, which is particularly relevant in Bangladesh given the impact of monsoon rains on driving behaviour.

3.2 Layer 2 — Edge Computing (Roadside Processing Units)

A Roadside Processing Unit (RPU) is deployed at each major intersection cluster. Each RPU is a ruggedised server or industrial-grade single-board computer (for example, an NVIDIA Jetson AGX Orin) housed in a weatherproof enclosure. The RPU processes data from nearby IoT devices in real time and makes sub-second decisions without waiting for cloud confirmation.

- **Local Signal Optimisation:** The RPU runs a lightweight reinforcement learning agent trained centrally and periodically updated via the cloud. Using live queue length and

speed data, it determines optimal green-phase durations for each signal arm and issues commands to signal controllers in under 100 milliseconds.

- Incident Detection: An on-device convolutional neural network (CNN) processes camera frames to detect anomalies — stationary vehicles in moving traffic, sudden speed drops, or unusual crowd gatherings — and flags these as potential incidents within two to three seconds of occurrence.
- Emergency Vehicle Detection: Acoustic sensors and camera-based licence plate recognition work together to identify emergency vehicles. When detected, the RPU immediately extends the green phase on the emergency vehicle's approach road and flushes cross-traffic to a red hold, creating a green corridor ahead.
- Local Data Buffering: Should the cloud connection drop, the RPU stores up to 72 hours of compressed telemetry locally. Once connectivity is restored, buffered data is uploaded in batches for retroactive analysis.

3.3 Layer 3 — Cloud Intelligence and Analytics

The cloud layer, hosted on Amazon Web Services (AWS), handles workloads that are computationally intensive, require large datasets, or produce insights over longer time horizons than the edge layer can manage. The cloud acts as the brain of the whole system, training and distributing updated models back to edge units on a nightly basis.

4. Cloud Infrastructure Design on AWS

4.1 Data Ingestion — Amazon Kinesis Data Streams

All edge nodes publish telemetry (vehicle counts, speeds, incident flags) to Amazon Kinesis Data Streams. Kinesis is purpose-built for high-throughput, real-time data ingestion and can handle hundreds of thousands of records per second, which is necessary when the system scales across multiple cities. Each city is assigned a dedicated shard group to provide logical isolation.

4.2 Stream Processing — Amazon Kinesis Data Analytics

Amazon Kinesis Data Analytics (now part of Amazon Managed Service for Apache Flink) processes the incoming streams to calculate rolling averages of traffic density, detect city-wide congestion patterns in near-real time, and generate alerts when thresholds are breached. Results are written back to a processed stream consumed by the dashboard and notification services.

4.3 Compute — Amazon EC2

Core application servers — including the fleet management API, the city-wide traffic optimisation service, and the model training pipelines — run on EC2 instances. Specifically:

- Model Training Cluster: A fleet of GPU-equipped instances (ml.g4dn.xlarge via Amazon SageMaker) runs nightly batch training jobs that consume the previous day's telemetry and produce updated signal optimisation models.
- Application Servers: Three m5.xlarge EC2 instances sit behind an Application Load Balancer, running the REST APIs consumed by the dashboard, mobile applications for commuters, and integration endpoints for emergency services.

- Auto Scaling: EC2 Auto Scaling Groups ensure that if any instance fails, replacements are launched automatically within minutes, maintaining the high-availability requirement.

4.4 Serverless Event Handling — AWS Lambda

Time-sensitive, low-frequency events — such as incident notifications, emergency vehicle alerts, and anomaly detections — are handled by Lambda functions rather than persistent servers. This is cost-effective because these events are sporadic, and Lambda charges only for actual compute time. A Lambda function triggered by Kinesis processes each incident flag, enriches it with location metadata from DynamoDB, and pushes a notification to the relevant emergency dispatch system via SNS.

4.5 Storage

- Amazon S3: Raw telemetry archives, camera snapshots flagged during incidents, and trained model artefacts are stored in S3. Lifecycle policies automatically transition data older than 90 days to S3 Glacier for cost-effective long-term retention. Approximate storage requirement: 50 TB active, 200 TB archived.
- Amazon DynamoDB: A NoSQL table holds the real-time state of every intersection — current phase, queue length, last-update timestamp, and active incident flags. DynamoDB's single-digit millisecond read/write latency makes it appropriate for this operational state store. Auto-scaling capacity units handle traffic spikes without manual intervention.
- Amazon RDS (PostgreSQL): Historical trip records, route planning data, and relational data for public transport scheduling are stored in a managed PostgreSQL cluster with Multi-AZ deployment for resilience.

4.6 Machine Learning — Amazon SageMaker

SageMaker provides the managed environment for training, evaluating, and deploying traffic prediction models. The platform handles infrastructure provisioning for training jobs, model versioning, and A/B deployment between old and new model versions. Trained models are exported and compressed for deployment to edge RPU.

4.7 Security Services

- AWS Identity and Access Management (IAM): Enforces least-privilege access. Each IoT device, edge node, and application service has a dedicated IAM role with only the permissions required for its function.
- AWS IoT Core with TLS 1.3: All device-to-cloud communication is authenticated via X.509 certificates and encrypted in transit using TLS 1.3, preventing eavesdropping or message injection attacks.
- AWS KMS (Key Management Service): Encryption at rest for S3, DynamoDB, and RDS using AES-256 keys managed through KMS. Key rotation is enforced annually.
- Amazon CloudTrail and GuardDuty: Full audit logging of all API activity and automated threat detection for suspicious access patterns.

5. Deployment Model and Service Classification

5.1 Hybrid Cloud Deployment

The ISTMS uses a hybrid cloud model. The edge RPU's represent the private/on-premises component, providing local autonomy and low latency, while AWS provides the public cloud component for large-scale analytics and central management. This hybrid arrangement is necessary because:

- Latency for signal decisions (under 100 ms) cannot be met via a round-trip to a remote cloud data centre given Bangladesh's current backbone network conditions.
- Regulatory requirements around traffic and public safety data may prohibit storage of certain data types on infrastructure outside Bangladesh. The hybrid model allows sensitive data to remain on-premises while anonymised aggregates are sent to the cloud.
- Operations can continue locally during cloud outages, ensuring that traffic signals do not revert to static timing simply because an internet link went down.

5.2 Service Model Classification

The system uses all three primary service models in different parts of the stack:

- IaaS (Infrastructure as a Service): EC2 instances and networking components give the development team full control over the operating environment for custom applications. This is appropriate where the application has specific OS-level requirements.
- PaaS (Platform as a Service): SageMaker, Kinesis Data Analytics, RDS, and Lambda abstract away underlying hardware management, allowing the data science and development teams to focus on model logic and application code rather than server maintenance.
- SaaS (Software as a Service): The commuter-facing mobile application and the emergency responder dashboard are delivered as managed software services that end users access without concern for the underlying infrastructure.

6. Artificial Intelligence and Machine Learning Models

6.1 Traffic Density Forecasting — LSTM Neural Network

A Long Short-Term Memory (LSTM) recurrent neural network is used for short-term traffic density forecasting. LSTM networks are well suited to time-series prediction because they can learn temporal dependencies spanning many time steps — a critical capability for traffic, where today's morning peak volume is influenced by yesterday's patterns, public holidays, and seasonal weather.

The model receives as input: hourly vehicle counts per lane for the past 48 hours, weather readings, day-of-week indicator, and public holiday flag. It outputs predicted vehicle counts per lane for the next one, three, and six hours. These predictions allow the system to pre-adjust signal timings before congestion materialises rather than reacting after it has formed.

Model performance target: Mean Absolute Percentage Error (MAPE) below 8% for the 1-hour forecast horizon, evaluated on a held-out test set from Dhaka intersection data.

6.2 Adaptive Signal Control — Deep Reinforcement Learning

A deep Q-network (DQN) agent is trained in a simulated road environment (using SUMO, the open-source urban mobility simulator) to learn a signal switching policy that minimises average queue length across all arms of an intersection. The state space includes current queue

lengths, phase elapsed time, and estimated vehicle arrival rates. The action space consists of discrete phase selection choices.

The trained policy network is exported and deployed on each RPU. Periodically, the cloud retraines the model on real-world data accumulated since the last update and pushes the improved weights to all edge nodes, enabling continuous improvement as the system accumulates traffic experience.

6.3 Incident Detection — Convolutional Neural Network

A compact CNN (based on MobileNetV3 for computational efficiency on edge hardware) processes camera frames at four frames per second to detect: stopped vehicles in flowing lanes, pedestrians on carriageways, debris on the road, and sudden queue formation. The network was pre-trained on the ImageNet dataset and fine-tuned on a traffic incident dataset assembled from public CCTV footage and synthetic augmentation.

When an incident is detected with confidence above 0.85, the RPU sends an incident record to the cloud. A Lambda function enriches the record with GPS coordinates, notifies nearby emergency services via SNS push notification, and logs the event for subsequent analysis.

6.4 Route Optimisation — Graph Neural Networks

The road network of a city is naturally represented as a graph, where intersections are nodes and road segments are edges. A Graph Neural Network (GNN) operates on this structure to produce city-wide route recommendations that account for current congestion, predicted future congestion along candidate routes, and the physical capacity of each road segment. Commuters access these recommendations via a mobile application, and public transport dispatch receives optimised route guidance for detouring around incidents.

7. Data Communication and Protocols

7.1 MQTT for IoT Device Communication

Message Queuing Telemetry Transport (MQTT) is the primary protocol for IoT device-to-edge communication. MQTT is lightweight, operates over TCP/IP, and is designed for constrained devices with limited processing power and intermittent connectivity — exactly the profile of roadside sensors and signal controllers. AWS IoT Core acts as the MQTT broker for device-to-cloud communication, with QoS Level 1 (at-least-once delivery) configured for all sensor telemetry topics.

7.2 HTTPS/REST for Application Integration

System-to-system integration — between the cloud application layer, emergency dispatch systems, and third-party navigation providers — uses standard HTTPS REST APIs with OAuth 2.0 authentication. This ensures broad compatibility with external systems operated by government agencies and private transport operators.

7.3 WebSocket for Real-Time Dashboard

The traffic management dashboard used by city traffic control centres receives real-time updates via WebSocket connections, allowing map displays to refresh without polling. The WebSocket server runs on EC2 behind an Application Load Balancer with sticky sessions.

8. Security, Privacy, and Compliance

8.1 End-to-End Encryption

All data in transit between IoT devices, edge nodes, and the cloud is encrypted using TLS 1.3. Data at rest in S3, DynamoDB, and RDS is encrypted using AES-256 with keys managed by AWS KMS. This satisfies the security requirements of the Bangladesh National ICT Policy and aligns with international ITS (Intelligent Transportation System) data security standards.

8.2 Privacy by Design

Video footage from CCTV cameras is processed on-device for vehicle detection and then discarded. Only anonymised vehicle count and speed metadata is forwarded to the edge and cloud layers. No facial recognition is performed, and no individual vehicle tracking data is stored beyond the operational window needed for signal optimisation (typically 30 minutes). This approach ensures compliance with privacy expectations while still providing the data quality needed for traffic management.

8.3 Access Control

Role-based access control (RBAC) governs who can view and modify system configuration. Traffic control operators can view dashboards and acknowledge incidents. System administrators can update model parameters and device firmware. Emergency responders have read-only access to incident notifications and route recommendations. No external party can issue commands to physical signal controllers without passing through the authenticated API layer.

8.4 Fault Tolerance and High Availability

Each component of the system is designed with failure in mind. Edge RPUs operate independently and maintain local signal control without cloud connectivity. EC2 instances run in Multi-AZ configurations across two AWS Availability Zones in the ap-southeast-1 (Singapore) region, which currently offers the best latency from Bangladesh among AWS regions. RDS runs a standby replica that promotes automatically if the primary fails. S3 inherently provides eleven nines of durability.

9. Addressing Conflicting Technical Requirements

9.1 Low Latency vs. Centralised Processing

The edge-cloud split directly resolves this tension. Signal control decisions that must happen within 100 milliseconds remain at the edge, where the processing is collocated with the sensors. Analytics that can tolerate a latency of minutes or hours — model retraining, trend analysis, long-term planning — happen in the cloud. The two layers complement each other rather than competing.

9.2 Data Sharing vs. User Privacy

Aggregated, anonymised data flows to the cloud and can be shared with third parties (navigation apps, researchers, government planners) under data sharing agreements. Individual vehicle trajectories and camera footage never leave the local RPU. This architecture means that the system can produce public value through shared data while protecting individual privacy.

9.3 High Accuracy vs. Computational Cost

The signal control model deployed on edge RPUs is a compressed version of the cloud-trained model, produced using knowledge distillation and quantisation (converting 32-bit floating point weights to 8-bit integers). This reduces model size by approximately 75% with a loss of less than 2% in prediction accuracy — a trade-off that is well within acceptable bounds for this application.

10. Handling Infrequent Challenges

10.1 Sensor Failures and Calibration Drift

Each RPU continuously monitors the health of its connected sensors. If a sensor reading deviates beyond a statistically expected range (detected using an isolation forest anomaly detector running on the RPU), it is flagged as potentially faulty. The RPU switches its signal control logic to a fallback mode that uses camera-only data while raising a maintenance alert to the cloud. A field engineering team is dispatched via a ticketing system integrated with AWS SNS. Sensors undergo scheduled calibration checks every 90 days.

10.2 Network Instability in Remote Areas

RPUs are provisioned with dual SIM cards from two different mobile network operators, enabling automatic failover if one carrier's signal is unavailable. Additionally, 72-hour local telemetry buffering ensures that data is not lost during extended outages. For locations where mobile coverage is extremely poor, directional 900 MHz radio links connect the RPU to the nearest connectivity hub.

10.3 Large-Scale Real-Time Streaming Data

The use of Amazon Kinesis with auto-scaling shards ensures that the ingestion pipeline can handle the peak load from a fully deployed multi-city network without manual intervention. Load testing, conducted before each city-level rollout, simulates 150% of expected peak throughput to verify headroom. Kinesis retains data for 24 hours, providing a buffer that absorbs temporary processing slowdowns without data loss.

11. Monthly Operational Cost Estimation (AWS Pricing Calculator)

The following cost estimate is based on a phased deployment covering two metropolitan areas (Dhaka and Chattogram) with a combined 200 active intersections and 1,000 connected IoT devices. Prices are based on the AWS ap-southeast-1 region and reflect on-demand pricing; Reserved Instance discounts (approximately 30%) would reduce costs further for a committed deployment.

AWS Service	Configuration	Est. Monthly Cost (USD)
Amazon EC2	3x m5.xlarge (app servers) + 2x m5.2xlarge (processing), On-Demand, Linux	\$420

AWS Service	Configuration	Est. Monthly Cost (USD)
Amazon SageMaker	Training: ml.g4dn.xlarge x 50 hrs/month; Endpoint: ml.m5.large x2	\$310
Amazon Kinesis Data Streams	10 shards, 24-hr retention, ~5 million records/day	\$185
Amazon Kinesis Data Analytics	4 KPIs continuous processing	\$210
AWS Lambda	~8 million invocations/month, avg 500ms, 256 MB memory	\$45
Amazon S3	50 TB active storage + 200 TB Glacier archive + data transfer	\$640
Amazon DynamoDB	On-demand mode, ~100 GB data, 50 million read/write units/month	\$175
Amazon RDS (PostgreSQL)	db.r6g.xlarge Multi-AZ, 500 GB SSD storage	\$380
AWS IoT Core	1,000 devices, 6 messages/device/min = ~260 million messages/month	\$130
Amazon CloudFront + ALB	Dashboard CDN, API load balancer	\$75
AWS CloudWatch + CloudTrail	Monitoring, logging, audit trail	\$60
AWS KMS + GuardDuty + WAF	Security services	\$90
Data Transfer (egress)	~5 TB/month outbound to edge nodes and mobile clients	\$150
TOTAL (estimated)		~\$2,870 / month

Note: This estimate covers cloud infrastructure only. Physical hardware (RPUs, cameras, sensors, signal controllers) involves one-time capital expenditure estimated separately at approximately \$8,000–\$12,000 per intersection for a full installation. Cloud costs will scale approximately linearly as additional cities are added to the platform.

12. Scalability and Fault Tolerance Strategy

The system is designed to scale from a pilot deployment of 20 intersections in a single district to a nation-wide network covering all divisional cities without architectural changes. Scalability is achieved through the following mechanisms:

- Kinesis shards scale horizontally in response to throughput increases, handled automatically by the Kinesis auto-scaling feature.
- EC2 Auto Scaling Groups expand the application server fleet during periods of high dashboard or API demand and contract during off-peak hours, maintaining cost proportionality with usage.
- DynamoDB on-demand capacity mode absorbs sudden spikes — for example, during a major road accident that triggers a city-wide surge of API calls — without pre-provisioning.

- Each RPU is an independent unit; adding a new intersection to the network requires deploying an additional RPU and registering it with AWS IoT Core, with no changes to the cloud architecture.
- A multi-region expansion to additional AWS regions can be accomplished using Route 53 latency-based routing combined with S3 Cross-Region Replication for data locality compliance.

13. Public Safety and Emergency Handling

Emergency response integration is one of the most impactful features of the ISTMS. The following automated pipeline handles an emergency from detection to resolution:

1. An ambulance or fire truck approaching an intersection is detected by the acoustic siren sensor or camera-based licence plate reader on the RPU. The vehicle's emergency status is confirmed within two seconds.
2. The RPU immediately activates a green corridor: it extends the green phase on the emergency vehicle's lane, holds all cross-traffic on red, and communicates this state change to adjacent RPUs along the likely route (determined from GPS data) so they can pre-clear their intersections in sequence.
3. A Lambda function publishes an incident record to the emergency dispatch SNS topic. The dispatch centre's CAD (Computer Aided Dispatch) system receives this record via API integration and updates the incident log automatically.
4. The ISTMS route optimisation service computes the fastest currently-clear route from the emergency vehicle's current position to the reported incident location, accounting for live congestion, and pushes this recommendation to the vehicle's on-board navigation system.
5. Once the emergency vehicle has cleared the affected intersections, the RPUs automatically return to their normal adaptive signal control mode.

This end-to-end pipeline is expected to reduce emergency vehicle travel times by 30–40% based on comparable deployments in cities such as Singapore and Kuala Lumpur.

14. Interdependence of System Components

The various subsystems of the ISTMS are tightly interdependent, and the overall system performance is limited by the weakest link in the chain:

- Traffic prediction accuracy depends on sensor data quality. If a significant fraction of sensors at a cluster are faulty, the LSTM model receives incomplete inputs and its forecasts degrade. This is mitigated by sensor health monitoring and the fallback data imputation logic described in Section 10.1.
- Signal optimisation depends on AI inference speed. The DQN agent must produce a phase decision within the available compute budget of the RPU. This is why model compression (Section 9.3) is not optional — it is a hard operational requirement.
- Emergency response depends on network reliability. If the RPU cannot communicate with adjacent nodes to propagate the green corridor, the emergency vehicle gains a green light only at its immediate intersection rather than across its entire route. Dual-SIM redundancy and the mesh communication protocol between adjacent RPUs address this risk.

- Route optimisation depends on the completeness of the city graph model. Unmapped roads or recently changed road layouts will cause the GNN to produce suboptimal routes. A map update pipeline — integrated with OpenStreetMap's change feed and supplemented by GPS trace analysis — keeps the graph current.

15. Conclusion

The Intelligent Smart Transportation and Traffic Management System described in this assignment represents a practical, phased, and financially grounded approach to solving one of Bangladesh's most persistent urban challenges. By combining IoT sensing at the ground level, real-time decision-making at the edge, and large-scale analytics in the cloud, the system addresses the fundamental limitations of existing static, manually operated traffic infrastructure.

The hybrid cloud architecture ensures low-latency performance where it matters — at the intersection level — while preserving the ability to perform the computationally intensive work of model training and city-wide optimisation in a managed cloud environment. The AWS cost estimate of approximately \$2,870 per month for a two-city deployment demonstrates that the solution is financially viable for a government-funded smart city initiative, particularly when weighed against the economic cost of congestion, which the World Bank has estimated to cost Dhaka alone over \$3.8 billion per year in lost productivity.

With careful phased rollout, stakeholder engagement from traffic police, emergency services, and commuters, and a commitment to continuous model improvement, the ISTMS can meaningfully transform road safety, commute times, and emergency response capability across Bangladesh's growing cities.

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